

Banking, tracking, and sharing of materials

Have you ever looked for that one sample—cell line, plasmid, oligo, or reagent—only to find it unlabeled, misplaced, degraded, or missing entirely? Losing track of materials wastes time, duplicates effort, and can set projects back by weeks or months.

Most of what follows is light: a handful of conventions you set once and let do the work: fixed identifiers, labels that point to the inventory instead of carrying detail, indexed boxes, a quick line in the log when something moves. None of it asks much in the moment, and each piece pays back the next time someone needs to find, share, or trust a sample. Skip them and the cost hides in plain sight: half an hour here, a vial thawed twice there, the irreplaceable sample no one can locate when it matters.

Risk assessment

▷ Wear gloves, safety glasses, lab coat



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Procedures

» Setting up the inventory system

- (1.) **Critical:** Let the inventory issue identifiers as consecutive numbers with a material-type prefix. When the system hands out the next free number, no one accidentally picks a used one. 

Hint: Use three- or four-digit numbers with zero-padding. Strings sort numerically in any spreadsheet (without padding, p10 falls before p9), and partially smudged labels are recoverable; a 3-character identifier in a 4-digit scheme is clearly missing a digit. In many labs, plasmid identifiers take a “p” prefix, while oligonucleotides (primers, synthetic nucleotides, etc.) start with an “o”. If initials are customary, append as “o0036_MaC”.

- (2.) Prioritize putting the identifier on the label, nothing else matters more. With no room for much detail, the inventory becomes the natural place to look and to update.
- (3.) Keep the inventory simple enough that adding an entry is faster than skipping it. Back it up and keep distributed copies; losing the inventory is far more expensive than preventing that loss.

Note: Make logging the cheap path: a barcode scan that auto-fills the entry, a one-tap mobile form. With the cheap action being the correct one, the chain-of-custody fills itself.

- (4.) **Optional:** For large inventories, barcode vials or boxes to speed up handling. 

- (5.) Index every storage box and maintain digital box maps. Number boxes consecutively.

Hint: Leave the top-left position empty in every box; the visual asymmetry catches a misoriented box quickly.

Quality assurance: Audit box maps periodically and resolve mismatches immediately. Protect high-value stocks. Use locked boxes or controlled access for irreplaceable or sensitive materials, and track who has access. 

- (6.) Agree on what “validated” means for each material type before adding it to the inventory. For example, mycoplasma-negative cell lines, sequence-confirmed plasmids, spectroscopically pure compounds.

» General procedure: Banking a new material

- (1.) Pull the next identifier from the inventory.
- (2.) Label the tube or container in water-resistant ink or on a cryo-safe label. Frozen plastic doesn’t hold ink or stickers well.
- (3.) Record all required metadata in the shared inventory at the time of banking. At a minimum: plate/well ID, vector, host, passage, selection conditions, date, phenotype notes, and analysis method.

Hint: For biological materials, include origin, host/strain, selection markers, passage or batch number, date, and validation results. For chemical materials, document date of synthesis, purity, analytical confirmation, and stability information.

- (4.) Prepare multiple vials or aliquots whenever feasible, while the material is healthy and not after it has aged or been stressed. Store at least one in a backup location such as a separate freezer, with a collaborator, or in a repository.
- (5.) Place the vials into indexed cryoboxes or storage racks and update the digital box map immediately.

Banking of continuously renewable materials

- Purified plasmids
 - Bacterial or yeast glycerol stocks
 - Standard oligonucleotides
- (1.) Maintain at least one backup aliquot for each material under long-term storage conditions. Keep prior batches for at least two years to enable troubleshooting and reproducibility checks.
 - (2.) Withdraw a sample from the remaining working stock for your experiments.
 - (3.) When renewing a stock, assign a new identifier and validate its quality before adding it to the inventory.

Banking of cell lines

- Immortalized cell lines (HeLa, MCF 10A, HEK293, etc.)

